

High-Quality Chinese Course Plan for a Personal Duolingo-Like App

Executive Summary

A strong Chinese course for a mobile-first, Duolingo-like app should **not** be a simple HSK word list wrapped in flashcards. The best reference products each contribute a different strength: Duolingo contributes path-based progression, frequent review, and high-volume bite-sized practice; HelloChinese contributes stronger explicit support for pronunciation, handwriting, native-video input, and spaced repetition; Integrated Chinese contributes clearer lesson architecture and cumulative review; the official HSK syllabus contributes the most defensible task, topic, vocabulary, character, and grammar scope; and ACTFL/CEFR add outcome-based “can do” benchmarks that are better for product design than test labels alone. The result should be a **communicative-first, literacy-supported, frequency-biased curriculum** that reaches **roughly HSK 1–4 equivalence**, with reading-character coverage grounded in the official HSK 2026 syllabus and performance targets framed with ACTFL/CEFR-style descriptors. ¹

For adult English-speaking beginners, the recommended structure is a **Starter band plus four HSK bands**, delivered as a **single guided path with optional side tracks**. The core path should prioritize listening and speaking early, then steadily increase reading, character recognition, sentence building, and short-form writing. A reasonable authored scope is about **186 core lessons** organized into **20 units**, with cumulative targets of **300 / 500 / 1,000 / 2,000 vocabulary items** and approximately **246 / 371 / 655 / 1,096 cumulative recognition characters** through the HSK 1–4 bands, using the official 2026 HSK syllabus appendices for vocabulary and recognitional character counts. Because HSK, CEFR, and ACTFL describe proficiency differently, the CEFR/ACTFL mapping in this report should be treated as approximate and product-facing rather than test-equivalence law. ²

The learning engine should use **short lessons, heavy retrieval practice**, and **adaptive spaced repetition**. The research base is clear that spaced practice outperforms massed practice for long-term retention, repeated retrieval improves delayed recall, and adaptive forgetting-curve models are appropriate for personalized scheduling. For this app, that means each lexical item should live in separate but linked memory tracks for **recognition, production, and character form**, with review intervals lengthening only after successful retrieval. Tone learning should receive dedicated practice because tonal perception is especially difficult for learners from non-tonal backgrounds, and character learning should emphasize radicals, low visual complexity first, and selective handwriting rather than exhaustive pen-and-paper writing of every new word. ³

The curriculum proposed here assumes **adult English speakers, Simplified Chinese first, Mainland Putonghua pronunciation, typing-first writing**, and **mobile-first delivery**. Unspecified dimensions that remain product choices are the degree of Traditional Chinese support, whether handwriting is optional or required at graduation, the amount of exam-prep content versus communicative content, and how much human editorial review is available for AI- or template-generated lesson materials. Those assumptions

matter because they affect pacing, asset production, review density, and the exact shape of assessment.

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Design Basis and Assumptions

The recommended course combines five source families: **official standards, high-performing commercial course patterns, major college-textbook architecture, proficiency frameworks, and SLA/cognitive-science findings**. The table below shows what to borrow from each reference and what not to copy blindly. The comparison synthesizes official descriptions from Duolingo, HelloChinese, the HSK 2026 syllabus, Integrated Chinese, LingoDeer, ACTFL, and CEFR. 5

Reference	What it officially emphasizes	What to borrow	What to compensate for
Duolingo	CEFR-aligned course structure in major courses, built-in speaking/listening growth, section-level grammar/vocabulary visibility, strong gamification and progress scoring	Guided path, short lessons, frequent review, score/progress feedback, high repetition density	Add more explicit grammar explanations, stronger tone/character instruction, clearer Chinese-specific literacy scaffolding
HelloChinese	Game-based curriculum, all-in-one skill mix, speech recognition, handwriting, native-speaker videos, SRS	Chinese-specific onboarding, tone work, handwriting input, native video comprehension, integrated SRS	Add more formal milestone testing and a more transparent can-do progression model
HSK 2026 syllabus	Task, topic, vocabulary, character, and grammar syllabi for each level	Use as the primary scope authority for HSK 1–4 equivalence	Do not let the course become test-only; add richer communicative tasks and app-friendly sequencing
Integrated Chinese	Four-volume progression from no prior experience to Intermediate High, pinyin/tone exercises, cumulative review after every five lessons	Clean unit architecture, explicit learning objectives, cultural modules, review cadence	Make it smaller-grain and more mobile-native; decrease text density
LingoDeer	Grammar/pronunciation guidance, native audio, bite-sized lessons, beginner-to-B1 ambition	Explicit grammar notes, practical audio-rich micro-lessons	Add stronger Chinese media/story content and more robust adaptive review

Reference	What it officially emphasizes	What to borrow	What to compensate for
ACTFL / CEFR	Can-do proficiency descriptors across communication modes and global level descriptions	Use outcome statements for placement, mastery, and graduation criteria	Avoid pretending the frameworks map perfectly to HSK levels

The key assumption in this report is that the app is **communicative by default but exam-compatible by design**. In practice, that means the HSK syllabus defines *scope*, while ACTFL/CEFR help define *what learners should be able to do*, and commercial app references help define *how the path feels*. That hybrid is more rigorous than copying any one source whole. ⁶

A second important assumption is **Simplified-first, Traditional-later**. The official HSK infrastructure is centered on Standard Chinese and exam-aligned syllabi, HelloChinese and most large beginner apps optimize for beginner-friendly onboarding, and mobile Chinese input methods make **typing** much more central to real-world use than extensive handwriting. For adult mobile learners, the best compromise is: recognize many characters, type many characters, handwrite a carefully chosen high-frequency subset, and treat handwriting fluency beyond that as an optional track. That recommendation is also consistent with recent reviews that show technology-enhanced Chinese character learning works best when orthography, meaning, sound, and component analysis are integrated rather than taught as isolated penmanship drills. ⁷

A third assumption is **frequency-first selection inside HSK bands**. HSK gives a defensible target inventory, but subtitle-based frequency data suggest that word and character frequencies derived from spoken-like media are better aligned with everyday language exposure than counts drawn only from written sources. For an app intended to feel useful early, the internal ordering of HSK vocabulary should therefore be biased toward the most frequent, reusable, and combinable items first, even when the official HSK list itself is not ordered that way. ⁸

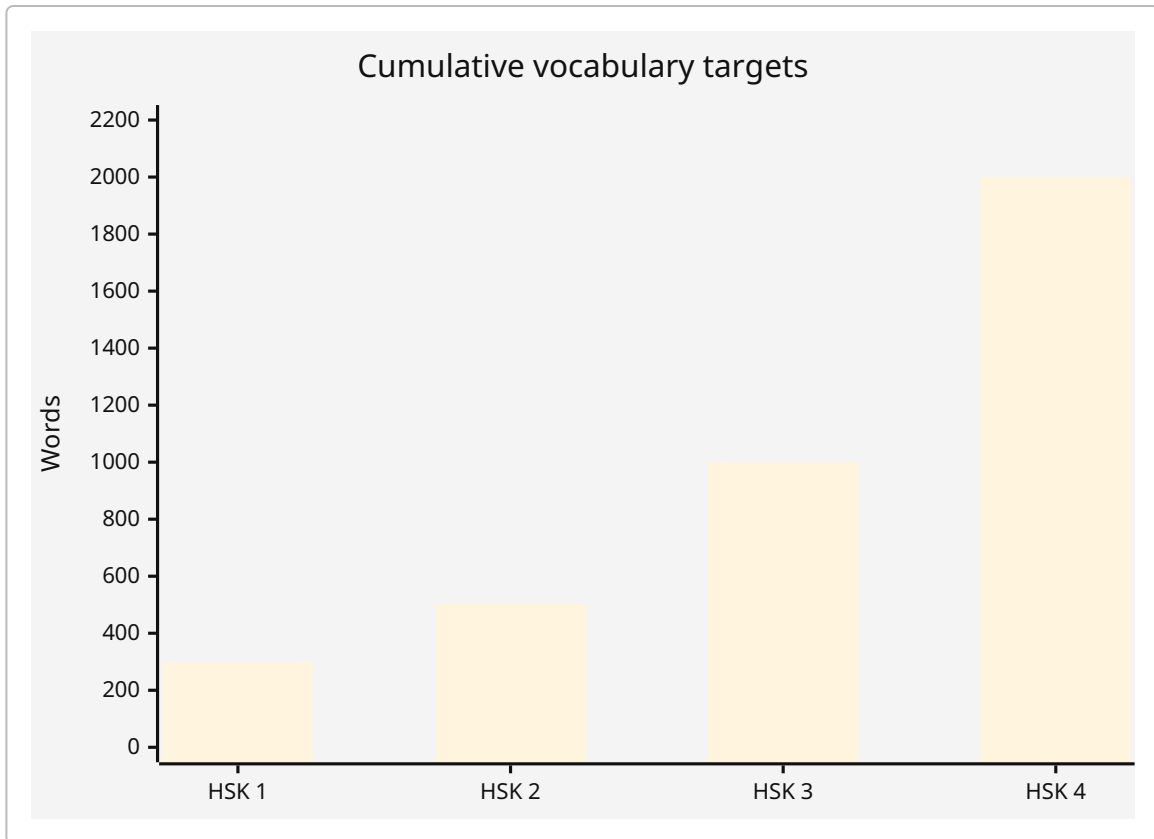
Curriculum Architecture

The course should be organized into a **Starter band** followed by **HSK 1, HSK 2, HSK 3, and HSK 4 equivalence bands**. The Starter band teaches pinyin, tones, Chinese input, and a very small set of survival utterances; it does **not** add extra off-syllabus burden, because its lexical content is drawn from the first HSK band. From a product perspective, this prevents the common beginner failure mode of being asked to memorize characters or translate sentences before learners understand the sound system. That sequencing is consistent with Chinese-specific app design patterns and with research showing that tone perception and character acquisition both benefit from careful front-loading of perceptual and structural support. ⁹

The band-level targets below combine official HSK cumulative vocabulary counts with cumulative recognition-character counts observable in the 2026 HSK syllabus appendices. The functional labels use approximate ACTFL and CEFR anchors. These mappings are intentionally approximate because HSK, ACTFL, and CEFR measure different things: one is an exam-centered Chinese syllabus, and the others are broader proficiency frameworks. ²

Band	Scope target	Cumulative vocabulary	Cumulative recognition characters	Product-facing proficiency target
Starter	Sound system, pinyin, tones, IME typing, survival classroom/ social phrases	included inside HSK 1	included inside HSK 1	Pre-Novice to Novice Low
HSK 1	Personal info, daily objects, food, simple routine, simple transactions	300	246	Novice Mid to early Novice High, early A1
HSK 2	Expanded daily life, directions, health, hobbies, simple comparisons, school/work basics	500	371	Novice High, A1 to A2 bridge
HSK 3	Familiar-topic conversation, experience, plans, opinions, community, travel, services	1,000	655	Intermediate Low, around A2
HSK 4	Explanation, narration, comparison, problem-solving, housing, work, society, culture, light media	2,000	1,096	Intermediate Mid, strong A2 to B1 bridge

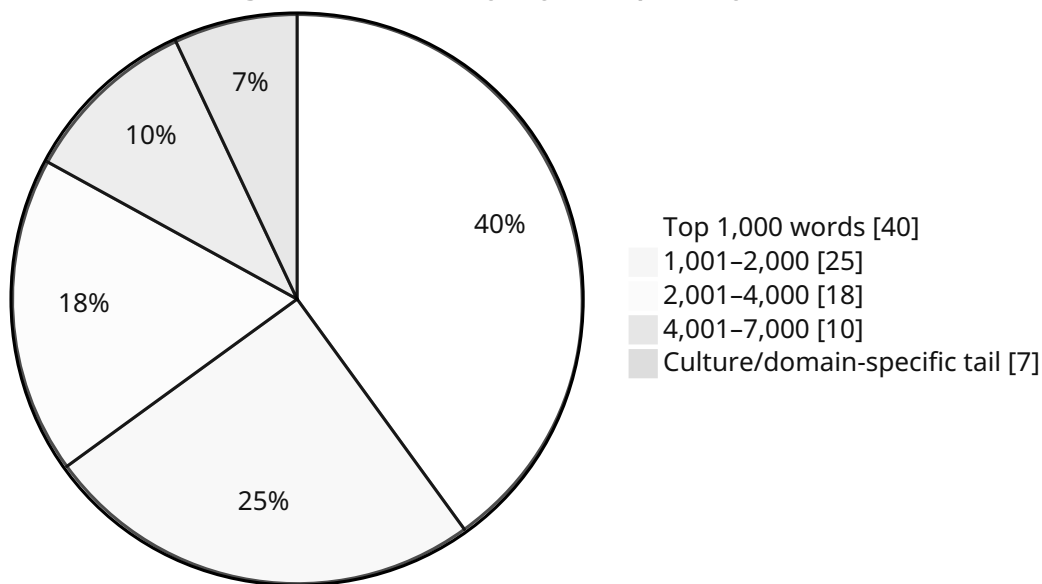
The cumulative HSK vocabulary targets can be visualized simply:



These counts come from the official 2026 HSK syllabus vocabulary appendices. ¹⁰

The internal vocabulary ordering should be explicitly **frequency-biased**, so that the taught lexicon tilts toward high-utility spoken words inside each HSK band rather than following bare alphabetical or list order. The chart below is therefore a **recommended design distribution**, not an official HSK tally. It is grounded in the HSK syllabus's daily-life task orientation and in SUBTLEX-CH's evidence that subtitle-derived frequencies are strong estimates of everyday exposure. ¹¹

Proposed share of taught vocabulary by frequency band



The proposed **20-unit skill tree** is below. It is a recommended synthesis built from official HSK topic/task bands and the strongest Chinese-learning design patterns from HelloChinese, Duolingo, Integrated Chinese, and LingoDeer. The grammar lists are selective rather than exhaustive; the official HSK grammar appendices should remain the source-of-truth tag set behind the scenes. ¹²

Unit	Band	Core communicative focus	Main grammar focus	Core vocabulary sets	Character focus
Starter A	Starter	pinyin, initials/ finals, tones, greetings, names	SVO basics; yes/ no with 吗	pronouns, greetings, country/name forms	你我好是吗 叫国
Starter B	Starter	numbers, classroom language, typing Chinese, simple identity	measure word 个; basic negation 不	numbers, age, nationality, basic nouns	人口女子 学生老师
Foundation A	HSK 1	self-introduction, family, occupations	是 sentences; possessive 的	family, student/ work roles	爸妈朋友 家人工作
Foundation B	HSK 1	dates, time, days, routine	time expressions; 在 for location/time	numbers, dates, daily routine verbs	年月日点 分时间起床

Unit	Band	Core communicative focus	Main grammar focus	Core vocabulary sets	Character focus
Foundation C	HSK 1	food, drink, ordering, likes/dislikes	想 / 要 / 喜欢; simple questions	meals, drinks, food, restaurant phrases	水饭茶菜 吃喝包饺
Foundation D	HSK 1	shopping, places, transport, weather	有 / 没有;多少钱; 这/那	stores, transport, prices, weather	钱买卖店 车天气冷 热
Daily Life A	HSK 2	home, location, directions, nearby places	里面/外面/上面; locatives	furniture, rooms, city places	房门楼路 旁边里外 上下
Daily Life B	HSK 2	health, feelings, invitations, hobbies	会 / 能 / 可以; comparisons with 比	body, sports, invitations, feelings	身体病医 药乐球快 慢
Daily Life C	HSK 2	travel planning, tickets, arriving/ leaving	来/去 compounds; direction complements	station, airport, travel logistics	机票站进 出回到离 近远
Daily Life D	HSK 2	school/work basics, simple opinion, simple narratives	因为...所以...; 虽然...但是...	class, office, simple abstract words	课教室考 试意思希 望准备
Community A	HSK 3	neighborhood, services, asking for help	existence and possession patterns; serial verbs	rent, community, service encounters	社区服务 帮助问题 需要
Community B	HSK 3	completed experience, past events, changes	了, 过, 着; result complements intro	travel, errands, experiences	经历完成 开始结束 变化
Community C	HSK 3	plans, intentions, arrangements	要/打算/准备; future time framing	scheduling, goals, appointments	计划安排 约定准时 迟到
Community D	HSK 3	preferences, reasons, soft opinion	比较; 觉得; 因为/ 所以; 还是	preferences, choices, reasons	选择理由 比较更最 觉得
Community E	HSK 3	media, rules, public behavior, culture entry	被动 intro; imperative/polite softening	signs, rules, sports, events, culture	规则比赛 新闻活动 文化

Unit	Band	Core communicative focus	Main grammar focus	Core vocabulary sets	Character focus
Independent A	HSK 4	housing, moving, repairs, utilities	把 construction intro; resultative verbs	housing, contracts, repairs	房租 搬修 理费 电水 厨房
Independent B	HSK 4	work life, meetings, applications, feedback	formal connectors; 把 / 被 expansions	office, CV, interview, meetings	办公 申 请 面 试 经 验 能 力
Independent C	HSK 4	health, technology, study tools, society	complements of degree; relative clauses intro	apps, devices, study/work tech	网 络 技 术 系 统 数 据 健 康
Independent D	HSK 4	storytelling, explanation, comparison, problem-solving	sequence markers; cause-effect; conditional forms	accidents, decisions, explanations	原 因 结 果 影 响 办 法 解 决
Independent E	HSK 4	culture, travel, media-lite, integrated tasks	discourse connectors; opinion support; summarizing	history-lite, customs, news-lite, travel contingencies	历 史 传 统 礼 节 旅 行 安 全

At the topic level, the HSK syllabus grows from **5/15/30 topic bands in HSK 1**, to **5/18/34 in HSK 2**, **6/22/54 in HSK 3**, and **7/31/77 in HSK 4**. That rise in topical breadth is one reason a course should not simply add more words per lesson at later bands; it also needs more varied dialogue situations, text types, and communicative tasks. ¹³

Sequencing, Review, and Lesson Mechanics

A good mobile Chinese course should be paced as a **steady weekly system**, not as an undifferentiated pile of lessons. The commercial references all emphasize short sessions and frequent practice, and meta-analytic evidence on mobile-assisted language learning supports app-based learning as an effective complement or primary environment when it is structured well. The recommended default is **5 new lessons per week**, plus **2 review blocks**, with lesson time increasing modestly as learners move from HSK 1 toward HSK 4. ¹⁴

Band	Recommended new lessons per week	Typical lesson length	Unit length	Review cadence
Starter	5	8–10 min	1–1.5 weeks	daily micro-review + weekly checkpoint

Band	Recommended new lessons per week	Typical lesson length	Unit length	Review cadence
HSK 1	5	8–12 min	~2 weeks	daily SRS + weekly review lesson
HSK 2	5	10–12 min	~2 weeks	daily SRS + biweekly cumulative review
HSK 3	4–5	12–14 min	2–2.5 weeks	daily SRS + review lesson every 4–5 lessons
HSK 4	4–5	12–15 min	2.5–3 weeks	daily SRS + cumulative review every 4–5 lessons

The lesson architecture should follow a consistent sequence: **preview → input → guided practice → productive practice → memory check → exit ticket**. That is much closer to what works in Chinese than “translation-only drilling.” The course should progressively increase the share of **typed and spoken production**, while keeping enough comprehension work to avoid overloading beginners. Tone and character work should be embedded rather than isolated, but both also need dedicated short drills. Official sources support more speaking/listening in modern app paths, all-in-one skill coverage, pinyin/tone exercises, and explicit speech-recognition or handwriting support. ¹⁵

The review system should be personalized at the item level. A practical engineering rule is to model recall as an exponential decay process and schedule review when projected recall probability falls below a threshold:

$$R(t) = e^{-t/S}$$

where **S** is item stability. The literature supports adaptive forgetting-curve models for language learning items and consistently favors spaced, retrieval-based practice over massed repetition for durable retention. For this app, the best default is to track separate stabilities for **recognition, production, and character form**, because learners often recognize a word long before they can say it or type it from memory. ¹⁶

The following **engineering defaults** are recommended as initial product parameters. These are **implementation proposals inferred from the literature**, not published constants from any one source, and should be tuned once learner telemetry exists. ¹⁷

Queue	First review after success	Next intervals if successful	Recall threshold to surface	Failure rule
Recognition	10 min	1d, 3d, 7d, 16d, 35d, 75d	0.82 early, 0.90 mature	reset to 10 min then 1d
Production	same session	1d, 2d, 5d, 12d, 28d, 60d	0.75 early, 0.88 mature	reset to same session then 1d
Character form	same session trace-copy-recall	1d, 4d, 10d, 21d, 45d	0.80 early, 0.90 mature	return to guided trace then recall

A robust *new-item* lesson should introduce roughly **6–10 new lexical items in Starter/HSK 1, 8–12 in HSK 2, 10–14 in HSK 3, and 12–16 in HSK 4**, but only when the lesson’s text remains mostly comprehensible. A safe rule is **80–90% known-word coverage** in comprehension passages and **70–80% known-word coverage** in production tasks. That balance keeps input readable while preserving enough novelty for growth. The recommendation follows app-style bite-sized design, HSK’s cumulative lexical growth, and research on desirable difficulty and retrieval-based learning. ¹⁸

The exercise system should be explicitly template-driven so that Cursor can generate materials at scale without losing curricular control. Each lesson should specify **objective tags, allowed vocabulary, required grammar, character whitelist, tone targets, error distractor rules, and known-word coverage limits**. Cursor then fills pre-approved exercise templates rather than inventing ad hoc pedagogy per lesson. That is the safest way to scale authoring while staying aligned with HSK scope and app-style consistency. ¹⁹

Skill	Exercise types	Template pattern	What Cursor supplies
Listening	hear-and-tap, picture match, select what you hear, short-dialogue comprehension	“Listen to A and choose the correct response”; “Match audio to scene”	dialogue scripts constrained to lesson vocab, distractors, audio scripts, comprehension questions
Speaking	repeat-after-model, minimal-pair production, cue response, role-play turn	“Say your name and nationality”; “Respond to a waiter”; “Produce Tone 2 vs Tone 3 contrast”	prompt lines, expected variants, STT intent lists, pronunciation target words
Reading	sentence-picture match, reorder sentence, short passage, story question	“Read and choose the correct room”; “Rebuild the sentence from tiles”	graded text with known-word thresholds, literal and inferential questions
Writing by typing	pinyin-to-hanzi, translation with whitelist, sentence completion, short free response	“Type: I want two cups of tea”; “Write a 25-character message”	accepted answers, alternate phrasing, IME hints, character whitelist
Character practice	radical identification, trace-copy-recall, component assembly, form discrimination	“Build the character from components”; “Which character matches the meaning?”	stroke-order data, radical labels, confusable distractors
Tone drills	contour recognition, minimal-pair listening, tone sandhi practice, phrase shadowing	“Choose mái vs mǎi”; “Shadow nǐ hǎo and lǎoshī”	minimal-pair lists, acoustic target info, phrase sets with third-tone and neutral-tone patterns
Grammar	pattern noticing, substitution, transformation, communicative completion	“Turn statement into question”; “Compare A and B with 比”	pattern frames, controlled substitutions, error distractors tied to English-speaker interference

For specific Chinese difficulties, the course needs *special* practice types, not just generic app exercises. Tone work should prioritize **perception before free production**, especially for English-speaking beginners, and should include minimal pairs, slow/clear model audio, and short phrase-level shadowing because novices from non-tonal backgrounds often struggle to perceive and categorize Mandarin lexical tones. Character work should prioritize **semantic radicals, low-complexity forms first, form-meaning-sound linking, and selective handwriting**, because characters with radicals and lower visual complexity are easier to acquire and because digital character pedagogy works best when feedback makes structural relations visible. ²⁰

Assessment and Progression

Assessment should be built around **placement, unit mastery, band milestones, and graduation**, not just around “did the learner finish the path.” The official HSK syllabus defines what each band should cover, while ACTFL Can-Do and CEFR help define performance across interpretive, interpersonal, and presentational modes. That means every major checkpoint should include at least some measure of **listening, reading, speaking/typed production, and character/orthographic control**. ²¹

A useful **placement test** has five short adaptive blocks: **sound system, listening, reading, sentence control, and production**. The sound-system block can be skipped for returning learners who place at HSK 2 or above, but it is valuable for adults who learned from pinyin-only resources and fossilized tone errors early. Placement should return both a **course start point** and a **mode profile**, so that learners can be placed, for example, at HSK 2 reading but HSK 1 speaking. Duolingo’s section visibility and score framing are good UI references for how placement results can remain motivating rather than discouraging. ²²

The recommended milestone structure is below. The “mastery” percentages are **product thresholds**, not claims about HSK pass scores. They are intentionally stricter than simple lesson completion because app graduation should mean durable, multi-skill control. ²³

Milestone	Evidence required	Pass standard
Unit checkpoint	12–18 items across listening, reading, controlled production, and 1–2 tone/character tasks	80% overall and no critical failure in listening
Band review gate	cumulative review on oldest and newest lesson sets	85% recognition retention on previously “mastered” items
HSK 1 graduation	self-introduction, basic transaction role-play, 300-word recognition coverage, basic typing	80% band exam, 85% retention on old items, 2-minute guided conversation
HSK 2 graduation	directions, health, plans, comparisons, 500-word recognition coverage	80% band exam, 70% productive control of high-frequency items, 3-minute guided interaction
HSK 3 graduation	direct exchange on familiar topics, simple narration, short typed note, 1,000-word recognition coverage	80% band exam, 75% on productive tasks, 80–120 character typed message

Milestone	Evidence required	Pass standard
HSK 4 graduation	narration/explanation, problem-solving, opinion support, 2,000-word recognition coverage	80% band exam, 70% productive control, 180–250 character typed response, 3–4 minute oral task

For **unit and band tests**, the item mix should change over time. In Starter and HSK 1, tests can be about **40% listening, 25% reading, 20% controlled production, and 15% tone/character**. By HSK 4, the mix should be closer to **30% listening, 25% reading, 25% production, 10% character/orthography, and 10% discourse-level grammar and synthesis**. This matches the increasingly varied task types and topical breadth in the official HSK syllabus and the broader communicative emphasis of ACTFL. ²⁴

The app should also include **micro-diagnostics** after every 4–5 lessons. Integrated Chinese’s cumulative review every five lessons is a particularly strong design cue to borrow here because it prevents the “I finished the lesson but forgot the unit” problem. In a mobile app, this should take the form of a **review node** that blends old and recent material, plus a short **mode-specific diagnostic** that can identify, for example, learners who read well but still guess tones poorly. ²⁵

Finally, it is worth separating **course graduation** from **exam preparation**. The core curriculum should target communicative HSK-equivalent ability. A later “test mode” can use official sample-question formats, time pressure, and exam strategy overlays. This separation protects the course from becoming too narrow too early, while still making exam alignment visible to users who need it. ²⁶

Content Operations and App Implementation

A realistic authored scope for a polished HSK 1–4-equivalent app is shown below. These are **planning estimates**, not immutable requirements, but they are sized to keep lessons small enough for mobile while still giving Chinese the extra density it needs for tones, characters, and controlled production. The totals are consistent with official HSK lexical growth and with the more complete, skill-rich lesson patterns visible in the strongest commercial references. ²⁷

Band	Units	Core lessons	Typical new words per lesson	Authored exercises per lesson	Checkpoints	Estimated new authored exercises
Starter	2	12	4–6	18–22	2	~240
HSK 1	4	32	6–10	20–22	4	~680
HSK 2	4	32	8–12	20–24	4	~700
HSK 3	5	50	10–14	22–24	5	~1,150
HSK 4	5	60	12–16	22–26	5	~1,440
Total	20	186	—	—	20	~4,210

That volume usually implies production of roughly **1,500–2,000 audio prompts, 250–350 short dialogues, 80–120 graded micro-readings, 40–60 story-style passages, a curated handwriting set for 200–300**

high-value characters, and a **tone minimal-pair bank** large enough to revisit common confusions repeatedly. Since HelloChinese highlights native-speaker videos and speech recognition, and LingoDeer emphasizes native-speaker audio, the strongest production standard is to use **human-recorded core audio for all high-frequency items and milestone content**, while letting high-quality TTS handle low-stakes variation once validated. ²⁸

A clean metadata schema is essential if Cursor is going to help author, transform, and QA lesson material. At minimum, every lesson should be a structured object rather than a prose file. A practical schema looks like this:

Field	Purpose
<code>lesson_id</code>	stable content identifier
<code>band</code> / <code>unit</code> / <code>path_position</code>	course routing
<code>can_do_targets</code>	learner outcomes in ACTFL/CEFR-style language
<code>hsk_scope</code>	HSK band and official-scope tags
<code>vocab_new</code> / <code>vocab_review</code>	lexical inventory control
<code>characters_new</code> / <code>characters_review</code>	orthographic scope
<code>grammar_tags</code>	syntax target list
<code>tone_focus</code>	minimal-pair and sandhi targets
<code>known_word_ratio</code>	reading/listening comprehensibility control
<code>exercise_mix</code>	percentages by task type
<code>media_refs</code>	audio/video/illustration assets
<code>accepted_answers</code>	production scoring support
<code>error_model</code>	distractor generation and feedback mapping
<code>accessibility_flags</code>	screen-reader notes, captions, high-contrast assets, font scaling checks
<code>review_policy</code>	queue parameters and thresholds
<code>version</code> / <code>editor_status</code>	publishing, QA, rollback

Cursor's role should be **structured generation, not unconstrained lesson invention**. In practice, Cursor should receive the lesson schema and then generate: a restricted dialogue script, comprehension items, distractors, answer variants, glosses, and short production prompts. It should also be used for **coverage checks**—for example, verifying that no untaught grammar slipped into a beginner dialogue, that distractors are plausible but not ambiguous, and that each lesson actually hits its objective tags. Human review is still necessary for tone-sensitive audio prompts, cultural nuance, register, and any free production scoring rubric. ²⁹

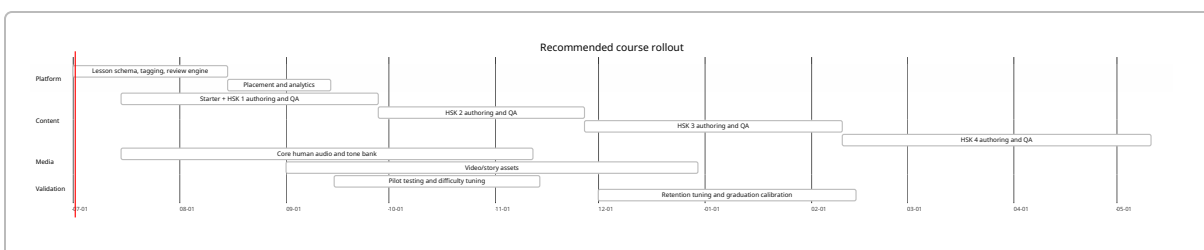
The path itself should be **branched but not fragmented**. The best structure is a required **Core Path** plus two optional tracks: a **Hanzi Track** and a **Challenge Track**. The Core Path guarantees communicative progression; the Hanzi Track adds extra component analysis, form recall, and handwriting for learners who want literacy depth; the Challenge Track adds freer production, cultural mini-projects, and slightly lower scaffold levels. Learners can step out into these branches but should reconverge at the same checkpoint nodes, so that the product still has a coherent mastery backbone. This is the cleanest compromise between Duolingo-style linear clarity and Chinese learners' uneven goals. 30

Adaptive difficulty should operate at three levels. First, **item level**: schedule reviews by projected recall probability and item difficulty. Second, **lesson level**: increase or reduce support, for example by toggling pinyin, slowing audio, or narrowing answer spaces. Third, **path level**: fast-track learners with strong receptive performance while still assigning extra tone or character work where diagnostics show weakness. Research on spaced practice, retrieval, and adaptive forgetting curves supports item-level personalization, and modern app products show that fine-grained progress feedback helps keep learners oriented. 31

Gamification should be present, but it should reward **retention and capability**, not just taps. Strong hooks include streaks, unit badges, daily quests, “tone streak” badges, radical-knowledge badges, XP for bringing old items back above retention threshold, and a visible progress score. Duolingo's score/progress framing is worth borrowing, but XP should be weighted toward successful retrieval of aged items and successful productive tasks rather than completion volume alone. Otherwise the app will optimize for activity instead of learning. 32

Accessibility should be designed in from the first lesson. At the UI layer, meet WCAG minimum contrast requirements, support Dynamic Type or equivalent text scaling, expose clear labels for screen readers, and ensure mobile tap targets and focus states are robust. At the pedagogy layer, **never encode tones by color alone**; also represent them with audio, contour icons, and text labels. Offer pinyin display modes, optional captions/transcripts, slower native audio, replay controls, stroke animations with pause/step controls, and a mode that privileges typing over handwriting for users with motor impairments. These choices align with W3C accessibility guidance, Apple and Android platform guidance, and systematic reviews of assistive/adaptive technologies in language learning. 33

A practical rollout plan is below. It assumes the app ships the memory model and authoring pipeline first, then gradually deepens content. This matters because Chinese content quality depends heavily on schema discipline and QA, not just raw lesson count.



Sample Beginner Syllabus

Below is a **10-week beginner syllabus** for the first part of the course. It assumes the pacing recommended earlier: 5 new lessons per week, short daily review, and one checkpoint at the end of each week. This

sample covers the Starter band and the first half of HSK 1. It is designed to get learners to the point where Chinese feels audible, typeable, and immediately usable, rather than abstract. The weekly themes reflect early HSK topics, Chinese-specific app onboarding needs, and beginner textbook sequencing. ³⁴

Week	Weekly goal	Sample lesson titles	Weekly checkpoint
Week One	Hear and produce the four tones; greet people; spell simple syllables in pinyin	"Meet Pinyin," "Tone One and Tone Four," "Hello and Goodbye," "What's Your Name?"	tone discrimination + greeting role-play
Week Two	Introduce yourself with name and nationality; use 吗 questions	"I Am from __," "Are You a Student?," "Chinese Input Basics," "Typing with Pinyin"	45-second self-introduction
Week Three	Use numbers, age, and measure words; identify people and classroom objects	"How Old Are You?," "One Teacher, Two Students," "This and That," "Who Is He?"	number dictation + simple Q&A
Week Four	Talk about family and basic occupations	"My Family," "Older Sister, Younger Brother," "My Mom Works Here," "Who Is in Your Family?"	family-tree speaking prompt
Week Five	Tell time and date; describe routine actions	"What Time Is It?," "Today, Tomorrow, Yesterday," "Get Up and Go," "Morning Class"	short routine timeline
Week Six	Order food and drink; say what you like and want	"Tea or Coffee?," "I Want Dumplings," "Do You Like Rice?," "At the Restaurant"	audio-based ordering task
Week Seven	Use 在 and 有 for place and existence; describe your room and neighborhood	"Where Is My Phone?," "There Is a Store Nearby," "My Room," "On the Left, On the Right"	map-following listening task
Week Eight	Shop, ask prices, and make simple choices	"How Much Is It?," "Too Expensive," "Which One Do You Want?," "At the Supermarket"	shopping dialogue
Week Nine	Travel locally, set simple plans, discuss weather and transport	"By Subway or Taxi?," "It's Cold Today," "Let's Go Tomorrow," "Where Is the Station?"	transit + weather mini-task
Week Ten	Consolidate core HSK 1 survival skills in one integrated scenario	"Welcome to Beijing Day One," "Meet a Friend," "Order Lunch," "Find the Hotel," "Weekend Review Project"	integrated beginner can-do assessment

By the end of this 10-week block, a learner on pace should be able to handle a small but meaningful set of beginner tasks: greet people, state basic personal information, count, tell time, order food, ask where things are, ask prices, make very simple plans, and type short high-frequency phrases in Chinese. That is a

better early success criterion for a mobile app than “finishing X vocab items,” because it aligns more closely with both HSK task framing and ACTFL-style can-do outcomes. ²³

Prioritized Sources

The most important sources for building this curriculum are the **official HSK 2026 syllabus and standards interfaces**, because they define the clearest public target for Chinese vocabulary, characters, topics, and grammar through HSK 4. For product-facing outcomes, the **2026 ACTFL Can-Do Statements** and the **Council of Europe CEFR level descriptions** are the next most important because they define what learners should be able to do across communication modes. ³⁵

Among curriculum and product references, the best primary sources are **Duolingo’s official course-and-score documentation**, **HelloChinese’s official feature and curriculum claims**, **Integrated Chinese’s official FAQ and sampler pages**, and **LingoDeer’s official Chinese course page**. Those sources are useful not because they are perfect models, but because they reveal which lesson structures, review expectations, and skill combinations successful products and programs consider essential. ³⁶

For learning science, the highest-priority papers are **Kim and Webb’s meta-analysis on spaced practice**, **Muqaibal, Kasprovicz, and Tissot on spacing in CALL vocabulary learning**, **Serfaty and Serrano on lag effects in vocabulary review**, **Zaidi et al. on adaptive forgetting curves**, and **Karpicke and Roediger on retrieval practice**. These are the strongest basis for the review engine and pacing policy in the course plan. ³⁷

For Chinese-specific pedagogy, the most important academic sources are **Pelzl’s review of why Mandarin tones are hard for L2 learners**, **Cao et al. on acoustic enhancement for tone learning**, **Kuo et al. on character properties in acquisition**, **Li’s review of technology-enhanced character teaching and learning**, **Guan et al. on handwriting effects**, and **Cai and Brysbaert’s SUBTLEX-CH frequency database**. Together, those sources justify the report’s emphasis on dedicated tone drills, radical-aware character instruction, selective handwriting, and frequency-first lexical ordering. ³⁸

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